

# Evanesca Technical Report: Open-Source Forensic Reconstruction of DeFi Value Extraction

Evanesca Project

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## Abstract

Evanesca is an open-source framework for reconstructing decentralized-finance (DeFi) transactions as Semantic Financial Graphs (SFGs) and auditing those graphs with transparent financial constraints. This report reorganizes the paper-era Evanesca research prototype as a public technical report and release document. The emphasis is forensic reconstruction, measurement, and analyst-facing evidence export, not real-time exploit detection or automatic attribution of malicious intent. We describe the threat model, financial semantics, SFG abstraction, constraint methodology, case-study evidence, paper-era measurement results, public artifacts, and remaining limitations. The current release includes source code, documentation, a JSON evidence schema, a replayed bZx case-study artifact, and public hash-set manifests.

## 1 Positioning

Evanesca should be read as a measurement and forensic reconstruction framework. It does not claim to be a production exploit detector, a low-false-positive monitor, or a system that infers malicious intent from traces alone. Its output is a candidate event: a transaction-local evidence trace that satisfies one or more financial constraints and therefore deserves analyst review.

This framing is important because value extraction in DeFi spans routine market activity and harmful protocol abuse. MEV and arbitrage can move value without a software defect [2, 8, 1], while flash-loan and oracle incidents can use similar swap, borrow, and repay primitives to extract value from a protocol-local economic assumption [6, 13]. Evanesca therefore separates reconstruction from final judgment:

- **candidate event:** a constraint-backed transaction or trace;
- **benign arbitrage:** value extraction that resolves price gaps without violating a protocol-local accounting, oracle, collateral, reward, bridge, or derivative-pricing rule;
- **harmful extraction:** abnormal value transfer caused by exploiting or violating such a protocol-local economic rule;
- **confirmed incident:** a publicly documented event with replayable on-chain transaction evidence and an external narrative.

The public release goal is narrower than a new conference submission. It is to make the system, evidence format, and reproducibility path inspectable: source code, technical report, documentation, reproducible artifacts, and example datasets should agree with each other and should avoid unsupported claims.

## 2 Motivation

DeFi attacks are often presented as isolated incidents, but many share recurring financial structure. A bZx-style price manipulation, a reward-distribution abuse, and a derivative exchange-rate defect may involve different contracts and event schemas, yet each can be inspected as an ordered sequence of financial actions with value flow, pricing dependencies, and accounting relations.

Existing systems address important parts of this problem. DeFiRanger and BlockEye focus on detecting known price-manipulation or attack patterns [9, 7]. DeFort and DeFiScope further analyze price manipulation with specialized logic [10, 11]. Operational systems such as Forta and OpenZeppelin Defender provide deployment-facing monitoring and alerting [3, 5]. These approaches are useful, but they tend to bind the analysis to a template, deployment rule, or specific attack class. Evanesca instead starts from the measurement question: what financial actions occurred, which service-class assumptions were stressed, and how did value move?

## 3 Threat Model

### 3.1 Adversary and Capabilities

Evanesca models transactions created by ordinary users, arbitrageurs, liquidators, MEV searchers, flash-loan users, and attackers. The sender may interact through one or more externally owned accounts or contracts. An adversarial transaction may compose multiple DeFi services atomically, borrow temporary capital, distort an AMM, oracle, lending market, reward pool, bridge, or derivative price, and then return most intermediate state to normal while retaining value.

The framework does not require private mempool access, block-builder control, or sandwich capability as core assumptions. Such behavior can be represented when visible in traces, but the core abstraction is transaction-local financial reconstruction.

### 3.2 Loss Subjects

The affected party depends on the service class. Evanesca distinguishes protocol reserves, liquidity providers, lending depositors, reward-pool participants, derivative-token holders, bridge backing reserves, traders receiving worse execution, and cases with no direct victim beyond market-efficiency arbitrage. This distinction prevents the system from treating every profitable trade as a harmful incident.

## 4 System Overview

Figure 1 shows the paper-era pipeline reused by the public release. The report reuses figures from the IMC, DeFi 26, and arXiv-era Evanesca drafts where they explain architecture or examples; empirical result tables in this report use the IMC measurement snapshot unless explicitly marked as validation evidence. The system has four stages.

**Semantic recovery.** Protocol-specific adapters decode traces and event logs into typed financial actions such as swap, deposit, withdraw, borrow, repay, liquidate, bridge, and reward. The adapter boundary is deliberate: protocol-specific decoding belongs in semantic recovery, while constraints should operate over service-class financial actions rather than incident-specific signatures.

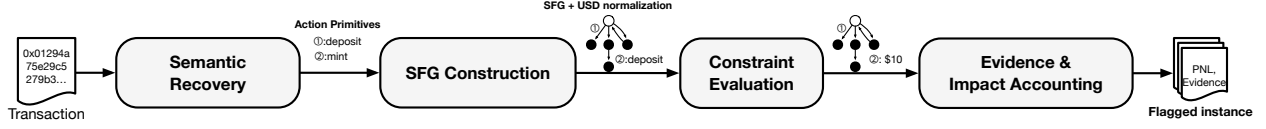


Figure 1: Evanesca pipeline: semantic recovery, SFG construction, constraint evaluation, and evidence export.

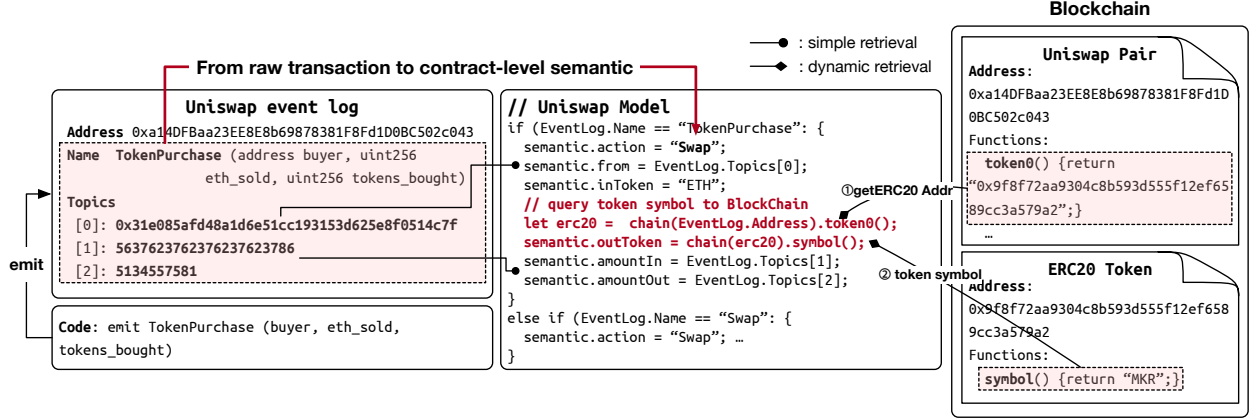


Figure 2: Semantic action model reused from paper-era Evanesca drafts. The figure documents the adapter boundary rather than adding a new experimental result.

**SFG construction.** Recovered actions are assembled into a directed graph  $G = (V, E)$ . Vertices represent accounts, contracts, pools, markets, services, and tokens. Edges represent typed financial actions and carry token amounts, USD-normalized values, protocol context, ordering, and raw evidence pointers. Raw transfers alone are not enough because many DeFi incidents depend on shares, collateral, debt, rewards, or derivative exchange rates that are not visible as a simple token-transfer chain.

**Constraint evaluation.** Constraints encode service-class financial assumptions: AMM value consistency, collateral sufficiency, oracle sanity, exchange-rate stability, reward allocation fairness, flash-loan structure, and bridge backing. A match is a candidate event, not an exploit verdict.

**Evidence export.** The public CLI exports JSON containing the transaction identity, summary, graph, constraints, evidence timeline, and diagnostics. The current schema is documented in `docs/public-api/analysis-result.schema.md`; example outputs live under `docs/example-datasets/`.

## 5 Semantic Financial Graph

The SFG abstraction is not claimed as a novel graph data structure by itself. Its contribution is the combination of reusable financial semantics, ordered evidence export, and constraint evaluation over cross-protocol actions.

Compared with cash-flow-only representations, Evanesca attempts to preserve financial relations that matter for DeFi analysis: pricing dependencies, collateral and debt relations, reward entitlement, derivative exchange-rate relations, bridge mint/redeem backing, action ordering, and PnL attribution.

Table 1: SFG elements exposed in the public evidence format.

| Element      | Public meaning                                                            |
|--------------|---------------------------------------------------------------------------|
| Node         | Account, protocol contract, pool, market, token, or service entity.       |
| Edge         | Typed financial action such as swap, borrow, repay, deposit, or transfer. |
| Value fields | Token amount, token symbol/address when known, and USD-normalized value.  |
| Ordering     | Transaction-local sequence used for evidence reconstruction.              |
| Diagnostics  | Unsupported actions, missing metadata, and recovery caveats.              |

## 6 Constraint Methodology

For an edge  $e$  with nonzero USD input value, Evanesca commonly computes:

$$\rho(e) = \frac{\text{valueUSD}(\text{out}(e))}{\text{valueUSD}(\text{in}(e))}.$$

The paper-era implementation used  $\alpha = 0.05$  for DEX swap edges and  $\beta = 0.02$  for lending or oracle-mediated stablecoin-derivative edges. The DEX threshold is above reported routine unexpected slippage on Uniswap [12]; the stablecoin-derivative threshold is tighter because oracle studies report smaller normal movements for major stable assets [4]. These thresholds are measurement parameters, not universal constants. A release-quality benchmark still needs sensitivity analysis and a public labeling protocol before making strong false-positive or false-negative claims.

Table 2: Current paper-era constraint families.

| Constraint family             | Service-class signal                                                               |
|-------------------------------|------------------------------------------------------------------------------------|
| PRICE_MANIPULATION            | Abnormal swap value ratio or price deviation.                                      |
| DEX_K_INVARIANT               | AMM reserve-invariant deviation or significant swap imbalance.                     |
| LENDING_COLLATERALIZATION     | Borrow, withdraw, or accounting behavior inconsistent with collateral assumptions. |
| EXCHANGE_RATE_MANIPULATION    | Vault or derivative exchange-rate anomaly.                                         |
| ORACLE_MANIPULATION           | Oracle-mediated price/accounting inconsistency.                                    |
| BRIDGE_INTEGRITY_VIOLATION    | Bridge backing or mint/redeem inconsistency.                                       |
| FLASH_LOAN_ATTACK             | Atomic temporary-capital pattern coupled with profit or manipulated state.         |
| CONCENTRATED_LIQUIDITY_ATTACK | Tick, liquidity, or price-impact anomaly in concentrated-liquidity markets.        |
| EMPTY_MARKET_ATTACK           | Low-liquidity lending-market share-price anomaly.                                  |

## 7 Case Study: bZx Replay

The public release currently includes one replayed case-study artifact: `docs/example-datasets/bzx-analysis-result.json`. It is generated by:

```
npm run analyze -- --tx \
0xb5c8bd9430b6cc87a0e2fe110ece6bf527fa4f170a4bc8cd032f768fc5219838 \
--chain ethereum --out docs/example-datasets/bzx-analysis-result.json --pretty
```

Figure 6 shows the paper-era bZx walkthrough used to motivate the pipeline. In the current public JSON export, the replayed transaction produces 10 graph nodes, 8 graph edges, 4 constraint records, and 8 evidence steps. The summary status is **flagged**; the current classifier leaves the category as **unattributed**. This is acceptable for a public schema example, but it also makes clear that the open-source release should not overstate the current classifier as a finished incident-labeling

Table 3: IMC scalar claims tracked for the public release.

| Quantity                        | Value      | Public interpretation                                               |
|---------------------------------|------------|---------------------------------------------------------------------|
| Ethereum transactions analyzed  | 14,187,803 | Research snapshot over supported DeFi interactions.                 |
| Confirmed incidents             | 40         | Curated replay set requiring explicit inclusion/exclusion criteria. |
| Total flagged instances         | 1,418      | Constraint-backed candidate events.                                 |
| Baseline arbitrage              | 956        | Benign baseline, not harmful by default.                            |
| Non-baseline instances          | 462        | Candidates requiring category-level review.                         |
| Implementation-defect instances | 219        | Reward-distribution plus derivative-pricing families.               |

## 8 IMC Measurement Snapshot

The following numbers summarize the IMC-version measurement snapshot. Other paper-era drafts are useful for figures and validation context, but this report does not import non-IMC scale-out result tables because some earlier drafts used broader claims that still need revalidation. Public materials must present these IMC numbers with scope qualifiers until the matching dataset manifests, population reproduction commands, and labeling protocol are complete.

The non-baseline breakdown is: 207 reward-distribution manipulation instances, 166 price-manipulation instances, 12 yield-bearing token deviation instances, and 77 unattributed long-tail candidates. The public report intentionally separates these from the 956 routine arbitrage baseline. Treating the baseline as false positives would misstate the measurement goal; treating all candidates as harmful would also be wrong.

### 8.1 IMC Case-Study Partition

The case-study partition follows the IMC version. Routine DEX arbitrage is reported as a benign baseline when the trace shows cross-market convergence without flash-loan, oracle, lending, reward, or derivative-pricing evidence. A candidate becomes non-baseline when the same value-ratio signal is coupled with state manipulation or cross-protocol dependency: price pumping, oracle-sensitive borrowing, reward trigger timing, or yield-token deposit/withdraw conversion. This distinction is why Figure 7 is central to the public report: it explains how Evanesca separates routine arbitrage from candidates that need deeper review.

### 8.2 Validation and Accuracy Evidence

The strongest accuracy-style evidence currently carried into this public report is the confirmed-incident validation from the paper-era regression suite: Evanesca reconstructs all 40 curated incidents under its reconstruction criteria. A prior scope-aware comparison table also reports DeFiRanger coverage of 33/34 within its price-manipulation scope, while Evanesca covers 40/40 across the broader incident set. This is validation evidence, not a full precision or recall claim for the 14.2M-transaction IMC scale-out dataset.

Table 4: Validation evidence retained for the public report.

| Evaluation item                    | Result       | Scope qualifier                                                                           |
|------------------------------------|--------------|-------------------------------------------------------------------------------------------|
| Confirmed-incident reconstruction  | 40/40        | Curated public-incident set; verifies reconstruction criteria, not population prevalence. |
| DeFiRanger within-scope comparison | 33/34        | Price-manipulation specialization from the paper-era comparison.                          |
| Baseline arbitrage audit           | 956 reviewed | IMC baseline separation; not a general false-positive-rate claim.                         |

Table 5: Current public artifact status.

| Artifact              | Status                                                                                                                          |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Source code           | Present with public CLI; repository-wide TypeScript build cleanup remains open.                                                 |
| Technical report      | This LaTeX/PDF report plus the Markdown planning report.                                                                        |
| Example datasets      | Synthetic schema sample and replayed bZx JSON.                                                                                  |
| Defect hash set       | 219-hash public-looking subset with manifest; provenance review required before tag.                                            |
| Non-baseline hash set | 1,386-hash broader candidate bundle with manifest; claim-scope review required before tag.                                      |
| GitHub release path   | Separate public repository at <a href="https://github.com/appff/Evanesca_public">https://github.com/appff/Evanesca_public</a> . |

## 9 Public Artifacts

The public release is organized around reproducible artifacts rather than paper-only claims.

The artifact validator checks the example dataset schema and the public hash-set counts:

```
npm run validate:public-artifacts
```

The release also contains a smoke-test GitHub Actions workflow that installs dependencies, validates artifacts, and checks the public CLI help path.

## 10 Reward and Derivative Findings

The paper-era investigation found two implementation-defect families that are kept as public hash-set artifacts after provenance review: reward-distribution manipulation and derivative-pricing/yield-bearing token deviation. Figure 8 illustrates a representative reward-distribution pattern. A flash-loan-funded swap triggers reward behavior inside token logic and the actor exits through a reverse path. The important point for Evanescence is not the specific token, but that the reward effect and the profit-taking path occur across contracts; a transaction-local SFG keeps those relations in one evidence trace.

Table 6: Scope comparison.

| System          | Primary input               | Main abstraction            | Public release distinction                                         |
|-----------------|-----------------------------|-----------------------------|--------------------------------------------------------------------|
| DeFiRanger [9]  | Traces/events               | Price-manipulation model    | Strong baseline for known price manipulation.                      |
| BlockEye [7]    | On-chain events             | Attack hunting/monitoring   | Operational attack hunting.                                        |
| DeFort [10]     | Transaction/state data      | Price-manipulation analysis | Specialized automatic PM detection.                                |
| DeFiScope [11]  | DeFi operations             | PM reasoning workflow       | Broad PM reasoning, LLM-assisted.                                  |
| Forta/OZ [3, 5] | Live deployment events      | Rule alerts                 | Real-time operational monitoring.                                  |
| Evanesca        | Traces and semantic actions | SFG plus constraints        | Reconstructs replayable candidate evidence across service classes. |

## 11 Comparison With Prior Work

Evanesca overlaps with prior DeFi attack-analysis and monitoring systems, but its public release targets a different operational point.

This comparison should be sharpened before any future conference submission: the public release can support case-by-case comparison, but it does not yet provide a complete ensemble baseline or a precision/recall table against all prior systems.

## 12 Reproducibility

A public claim should identify the code revision, configuration, dependency lockfile, environment variables, input transaction hashes or dataset manifest, command line, expected output, and limitations. The current minimum public workflow is:

```
npm install
npm run analyze -- --help
npm run analyze -- --tx <transaction-hash> --chain ethereum \
  --out out/analysis.json --pretty
npm run validate:public-artifacts
```

The bZx example has been replayed from a fresh public export without a local `cache/` directory, using public RPC fallback providers. Population artifacts still require external services such as Etherscan and Ethereum archive/RPC access, as documented in the artifact manifests.

## 13 Limitations and Release Gates

Evanesca remains a research prototype. Current limitations include incomplete adapter coverage, trace/RPC dependence, USD-normalization sensitivity, possible false positives from unusual benign

behavior, false negatives for unsupported protocols or hidden state updates, and incomplete public labeling for the full candidate population.

Before a stronger release tag or future conference submission, the project should complete:

- repository-wide TypeScript build cleanup;
- dependency audit triage;
- implementation-synced constraint predicate documentation;
- public inclusion/exclusion criteria for confirmed incidents;
- threshold sensitivity analysis for  $\alpha$  and  $\beta$ ;
- human-labeling protocol for precision/recall claims;
- additional curated case-study JSON exports beyond bZx.

## 14 Conclusion

Evanescence exposes DeFi transactions as replayable financial-action evidence rather than opaque alerts. The open-source release should therefore be judged by whether its code, report, schema, artifacts, and examples let another researcher inspect how a candidate was reconstructed. The current public package provides that baseline: a source tree, technical report, evidence schema, replayed bZx example, artifact manifests, and release checks. Future conference work can build on this foundation by adding stronger correctness evaluation, prior-work comparisons, and sensitivity analysis.

## References

- [1] Crypto Briefing. What is MEV? ethereum’s invisible tax explained. <https://cryptobriefing.com/what-is-mev-ethereums-invisible-tax-explained/>, 2021.
- [2] Flashbots. Flashbots: Illuminating the dark forest. <https://www.flashbots.net/>, 2024.
- [3] Forta Foundation. Forta: A decentralized runtime security solution for automated threat detection and prevention. <https://docs.forta.network/>, 2022.
- [4] Bowen Liu, Pawel Szalachowski, and Jianying Zhou. A first look into DeFi oracles. In *IEEE International Conference on Decentralized Applications and Infrastructures*, pages 39–48. IEEE, 2021.
- [5] OpenZeppelin. OpenZeppelin defender: Smart contract monitoring with rule-based sentinels. <https://docs.openzeppelin.com/defender/module/monitor>, 2024.
- [6] Kaihua Qin, Liyi Zhou, Benjamin Livshits, and Arthur Gervais. Attacking the DeFi ecosystem with flash loans for fun and profit. In *International Conference on Financial Cryptography and Data Security*, pages 3–32. Springer, 2021.
- [7] Bin Wang, Han Liu, Chao Liu, Zhiqiang Yang, Qian Ren, Huixuan Zheng, and Hong Lei. BLOCKEYE: Hunting for DeFi attacks on blockchain. In *IEEE/ACM International Conference on Software Engineering: Companion Proceedings*, pages 17–20. IEEE, 2021.



- [8] Ben Weintraub, Christof Ferreira Torres, Cristina Nita-Rotaru, and Radu State. A flash(bot) in the pan: Measuring maximal extractable value in private pools. In *Proceedings of the ACM Internet Measurement Conference*, pages 458–471. ACM, 2022.
- [9] Siwei Wu, Zhou Yu, Dabao Wang, Yajin Zhou, Lei Wu, Haoyu Wang, and Xingliang Yuan. DeFiRanger: Detecting DeFi price manipulation attacks. *IEEE Transactions on Dependable and Secure Computing*, 21(4):4147–4161, 2024.
- [10] Maoyi Xie, Ming Hu, Ziqiao Kong, Cen Zhang, Yebo Feng, Haijun Wang, Yue Xue, Hao Zhang, Ye Liu, and Yang Liu. DeFort: Automatic detection and analysis of price manipulation attacks in DeFi applications. In *Proceedings of the ACM SIGSOFT International Symposium on Software Testing and Analysis*, pages 1517–1529. ACM, 2024.
- [11] Juantao Zhong, Daoyuan Wu, Ye Liu, Maoyi Xie, Yang Liu, Yi Li, and Ning Liu. DeFiScope: Detecting various DeFi price manipulations with LLM reasoning. *IEEE Transactions on Dependable and Secure Computing*, 2025.
- [12] Liyi Zhou, Kaihua Qin, Christof Ferreira Torres, Duc Viet Le, and Arthur Gervais. High-frequency trading on decentralized on-chain exchanges. In *IEEE Symposium on Security and Privacy*, pages 428–445. IEEE, 2021.
- [13] Liyi Zhou, Xihan Xiong, Jens Ernstberger, Stefanos Chaliasos, Zhipeng Wang, Ye Wang, Kaihua Qin, Roger Wattenhofer, Dawn Song, and Arthur Gervais. Sok: Decentralized finance (DeFi) attacks. In *IEEE Symposium on Security and Privacy*, pages 2444–2461. IEEE, 2023.

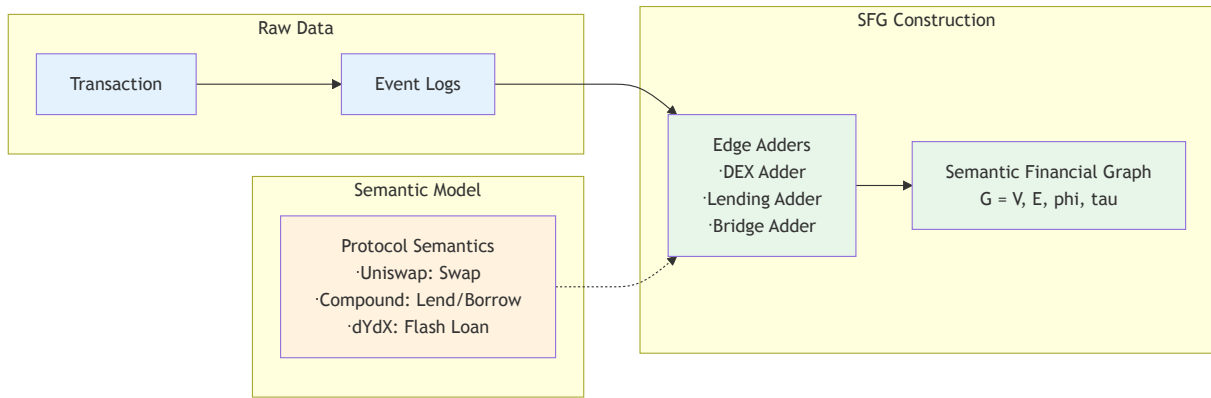


Figure 3: SFG construction figure reused from paper-era Evanesca versions.

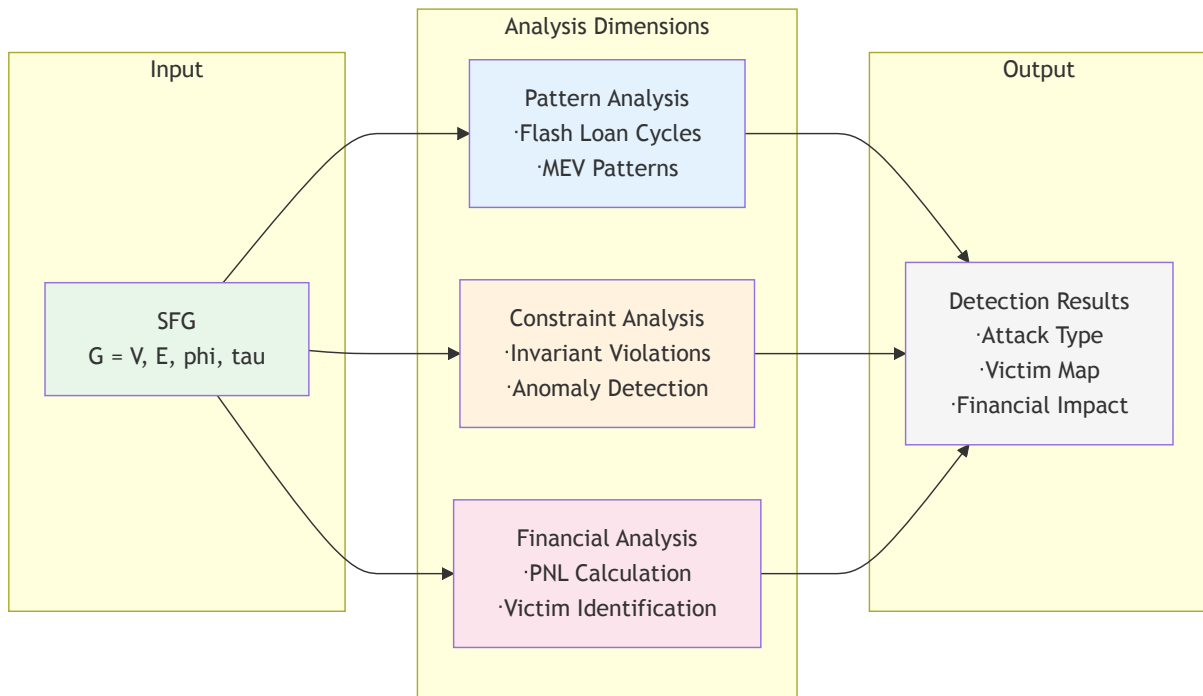


Figure 4: SFG evaluation figure reused to explain how graph evidence feeds constraint checks.

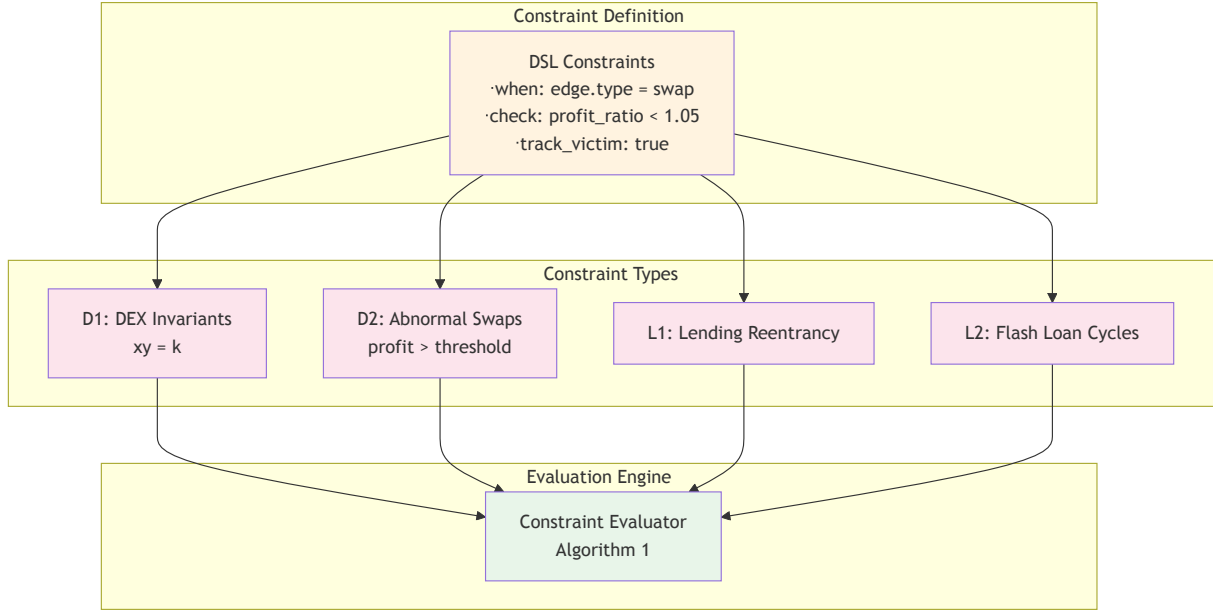


Figure 5: DSL constraint overview from the paper-era report. It is included as method documentation; the numeric evaluation below remains the IMC snapshot.

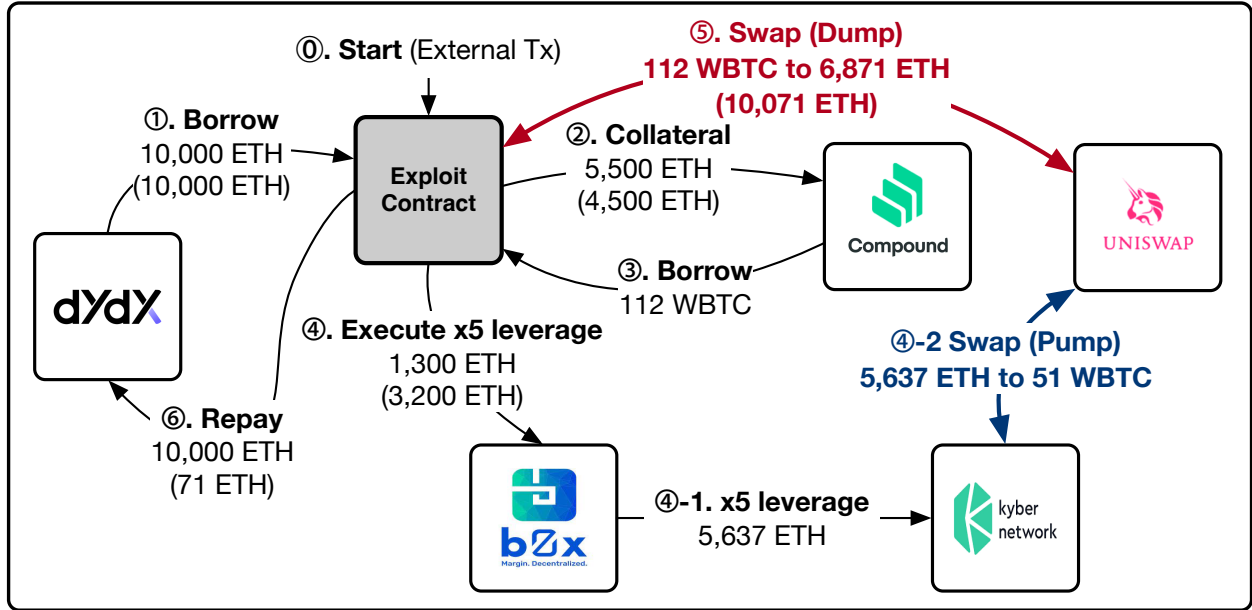


Figure 6: Representative bZx-style price-manipulation pattern from the paper-era walkthrough.

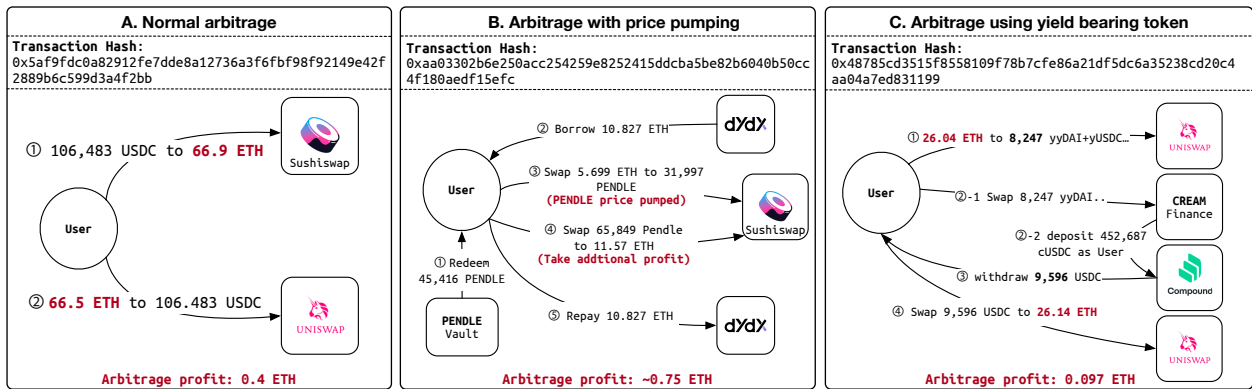


Figure 7: Representative evidence-trace patterns. Routine arbitrage is kept as a baseline, while manipulation and derivative paths require additional classification.

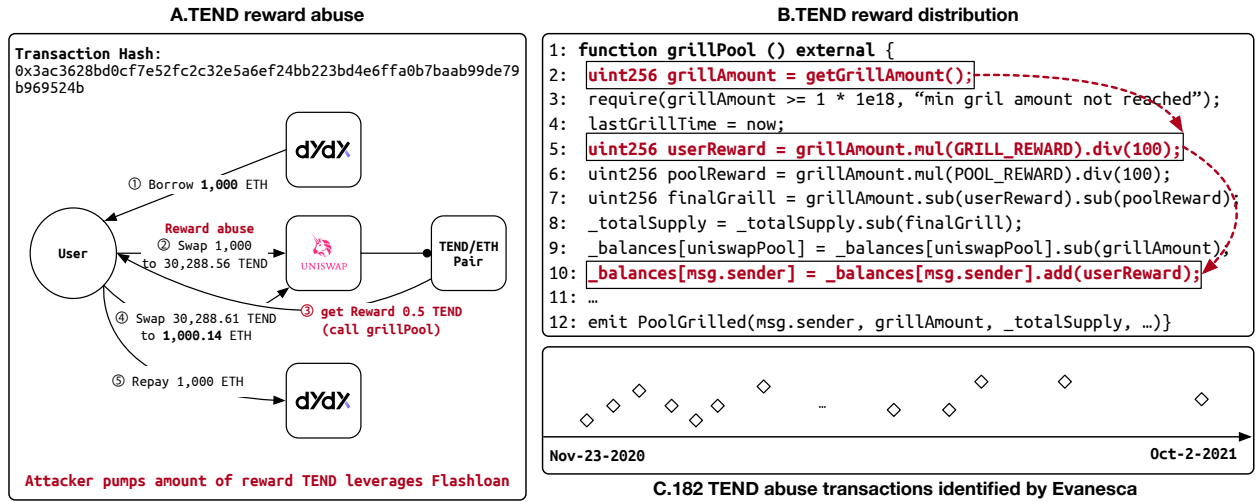


Figure 8: Representative reward-distribution manipulation pattern from the paper-era artifact set.